

COULD OUR
WASTE BE AN
ENERGY SOURCE?

ABSOLUTELY!!!
PRESENTING YOU
WASTE2**WATT**



SCENARIO ANALYSIS

THE STATE OF ENERGY



ENOUGH

Energy is in **scarcity** due to macroeconomic circumstances



ECONOMICAL

Rising prices of Natural gas, Electricity & Other Energy Sources



ENVIRONMENTAL

80% global energy consumption is via **fossil fuels**



EQUITABLE

Energy is an **inaccessible commodity** to many

ALTERNATIVE EVALUATION	Fossil Fuel	Solar Energy	Hydro Power	Wind Energy	Nuclear Energy	Biomass Energy*
Enough						
Equitable						
Environmental						
Economical						

More about
Biomass Energy



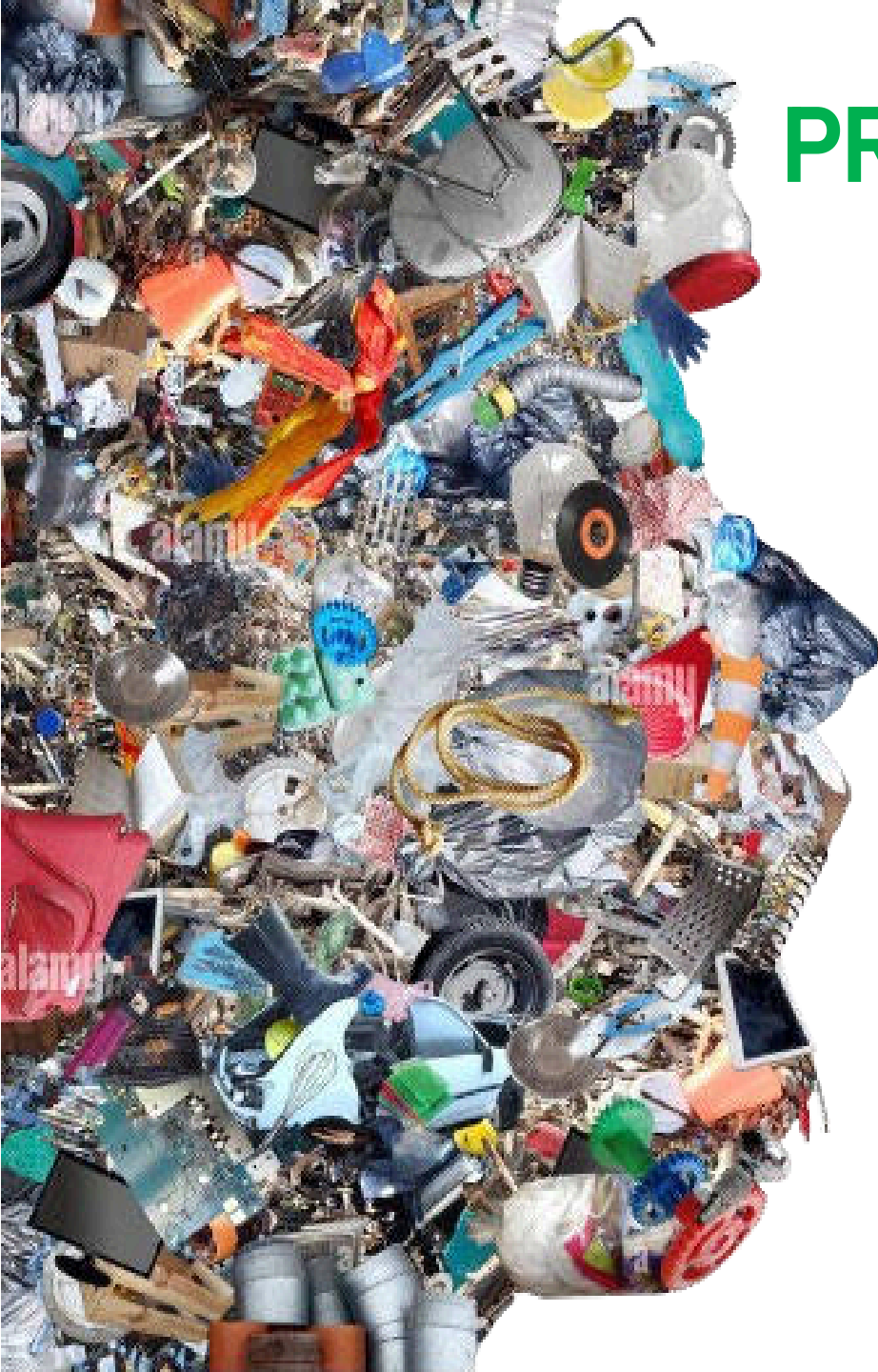
Input: Organic, Compostable, Food Waste



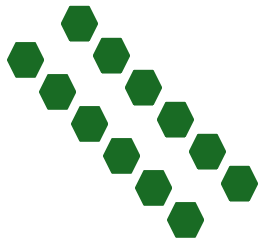
Potential Power Generation
1500 MWe



Potential alt. of LPG market
(14.41 lakh ton)



PROBLEM STATEMENT



Municipal
Organic Waste

TRASH TO TREASURE

Packaged
Bio-Fuel

THERE'S A LOT OF
WASTE IN DHAKA



6,500 Tons
waste Dumped
daily in Dhaka

IT'S POORLY
MANAGED



55% Uncollected
Urban Waste

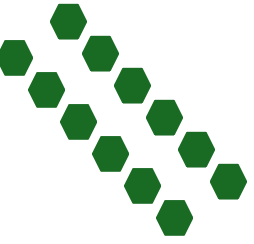


2x every 15 years
Waste volume

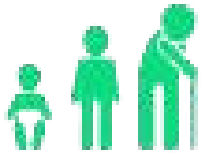


33-77% efficiency
waste collection

TARGET GROUP ANALYSIS



DEMOGRAPHIC



32-55



SEC C1
B2 B1 A3

GEOGRAPHIC



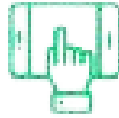
Urban
Suburban

VALUES & LIFESTYLE

BEHAVIORAL



Purchase Frequency
Monthly/ Bi-Monthly



Usage Frequency:
Daily/Frequently



Benefits Sought:
Effectiveness, Safety



Household Decision
Maker/Homemaker



Savings
Mindset



Seeks
Convenience



Cares for
Planet





BIG IDEA



INTRODUCING

WASTE²WATT

RENEWABLE BIO-GAS CYLINDERS

“Get juiced up from your waste”

29%
Cost
Saving

80%
Less
Carbon



Can be used for
**Cooking &
electricity gen**

CRUCIAL ROLE IN **ENERGY CRISIS**

BIO-FUEL ACROSS THE WORLD



ATEC BioDigesters

Turning compostable
waste into energy
powering day-to-day
agriculture activities

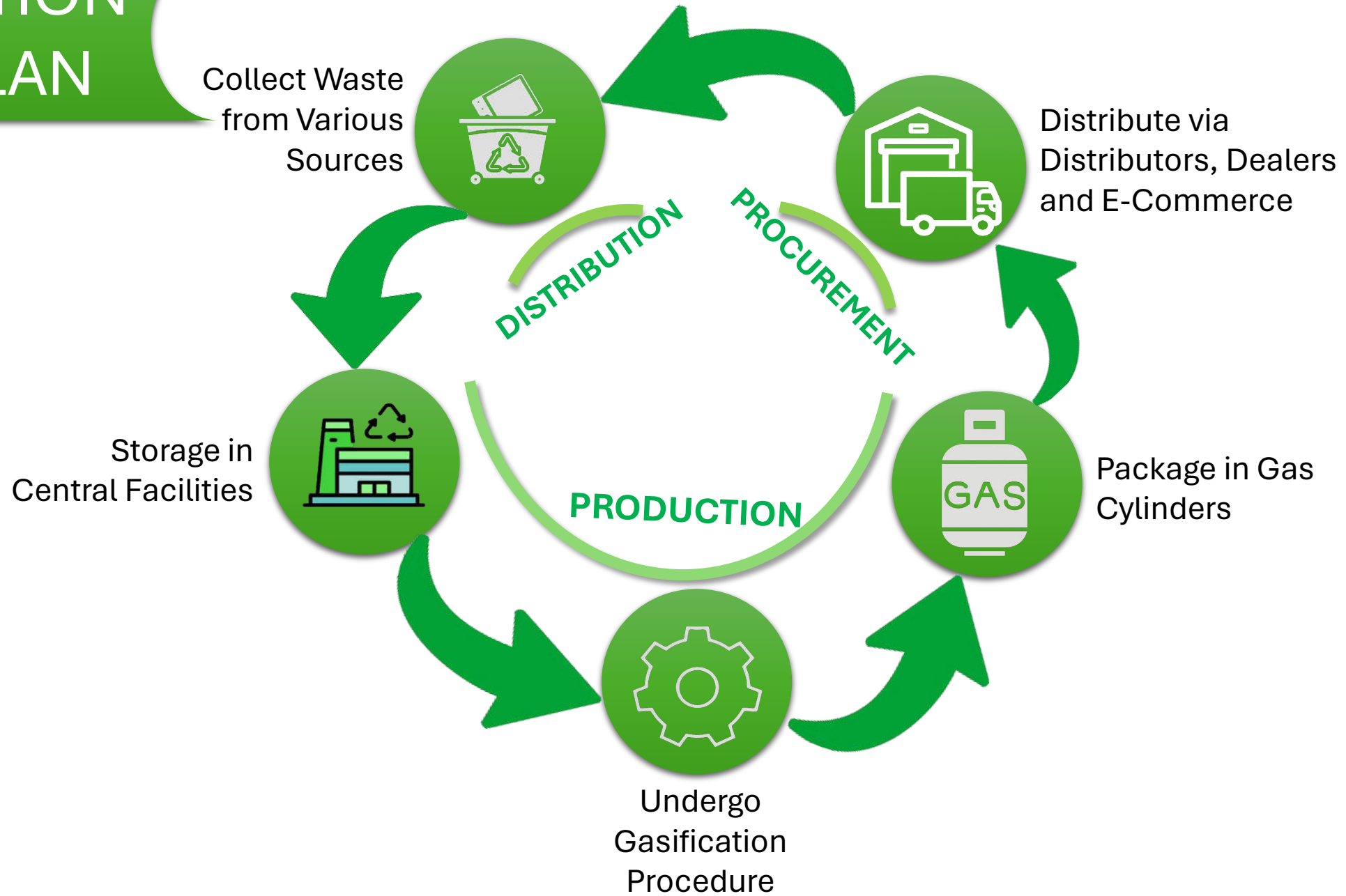


Husk Power System

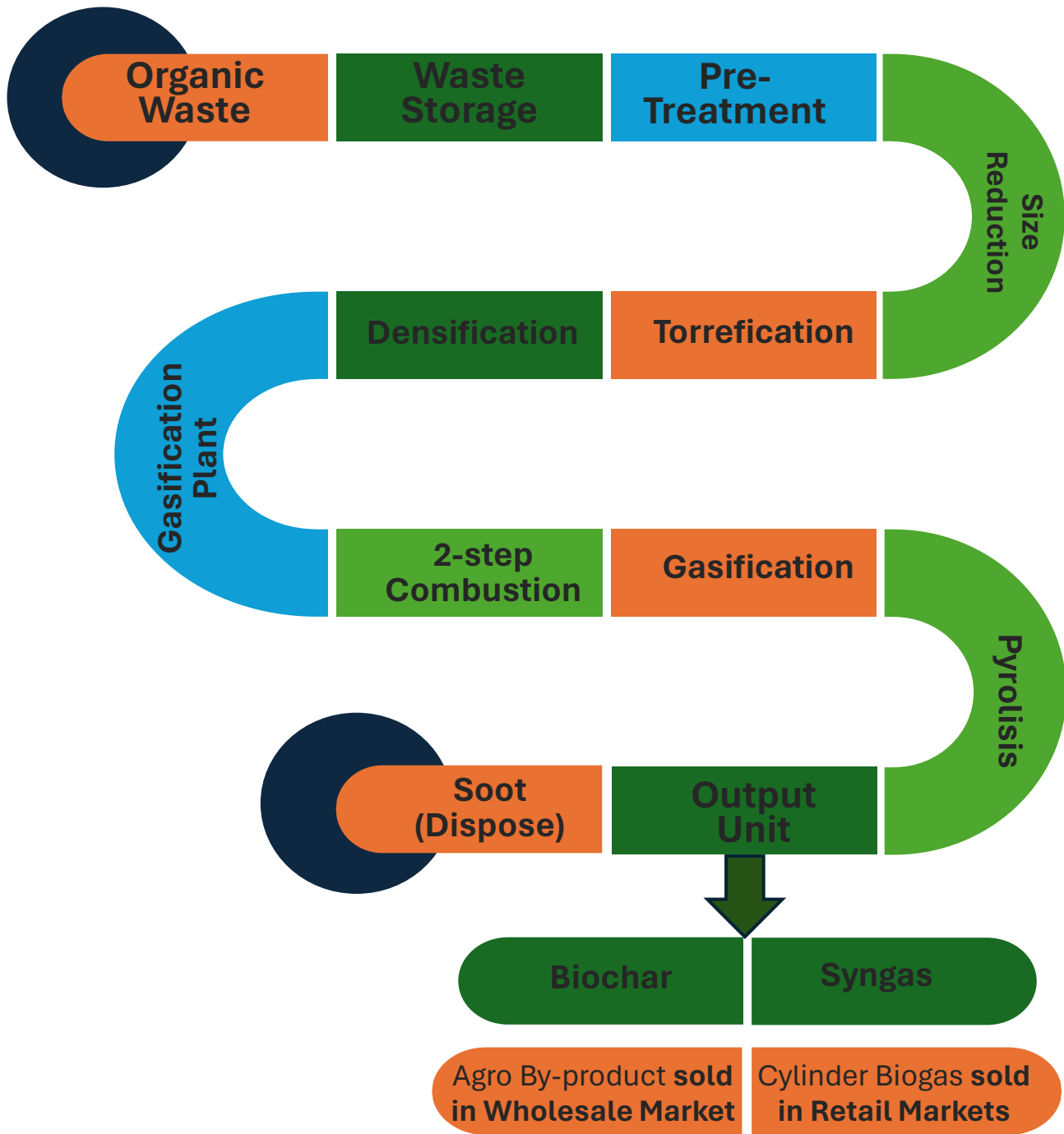
Turning agro-waste
into energy powering
rural village &
business + creating
employment



ACTION PLAN

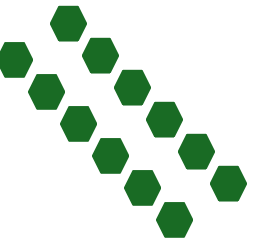


PRODUCTION PROCESS



ACTION
PLAN

MARKETING AND CAMPAIGN ACTIVITY



**Gas Cylinder
Dealers**

18,000 dealer
network within
Dhaka Metro City



E-Commerce

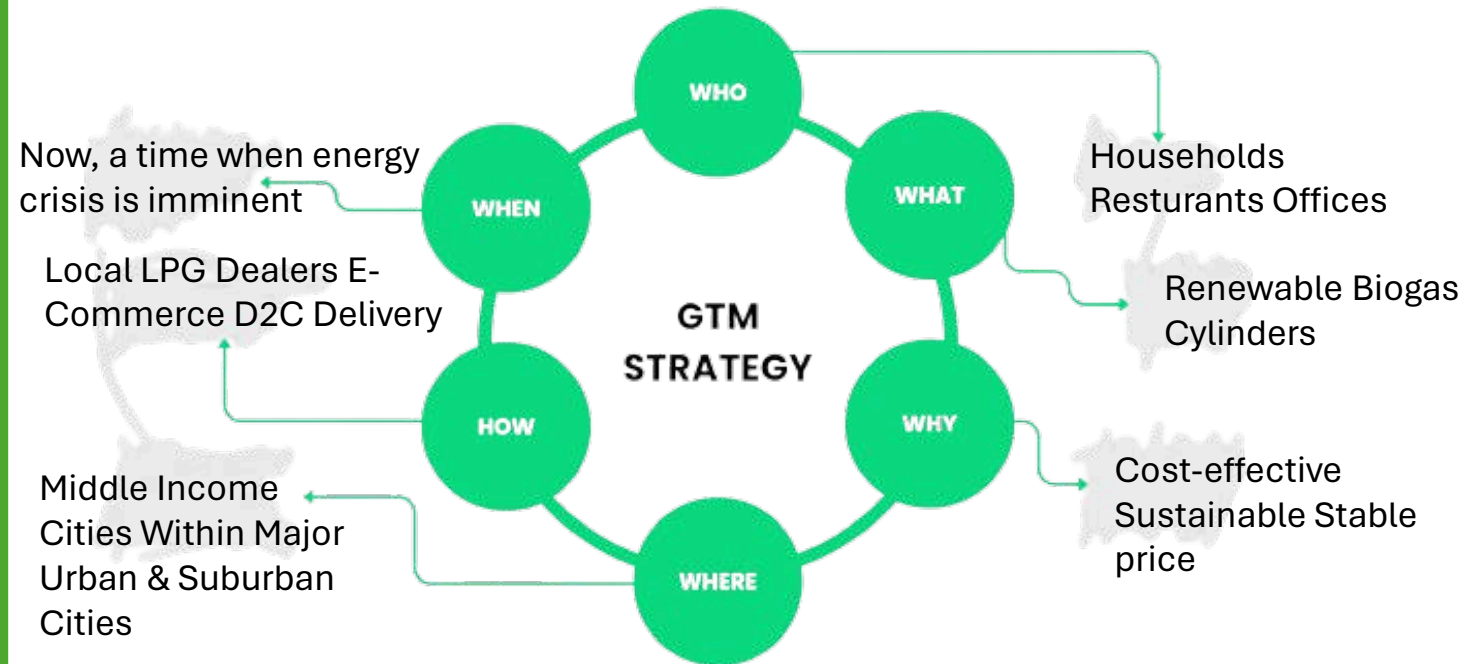
Sheba.xyz
Gasmonkey
Cylinderwala



**Direct to
Consumer**

Authorised
Distributors

Distribution Point



MARKETING AND CAMPAIGN ACTIVITY



PROCUREMENT

Initial Partners are businesses that produce high waste / unit

- RESTAURANTS
- HOTELS
- OFFICES
- HOSTELS
- WEDDING HALLS
- WET MARKETS
- RECYCLING PARTNERS

Pipeline Expanding into Sourcing from Direct Consumers like Households



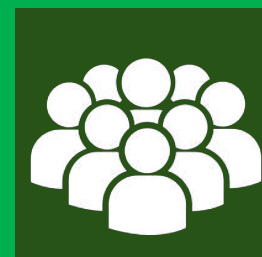
ESG IMPACT



ENVIRONMENT

Fossil Fuel Dependence Reduced

Organic Waste Management



SOCIAL

Awareness for Renewable Energy

Contributes sustainable city development



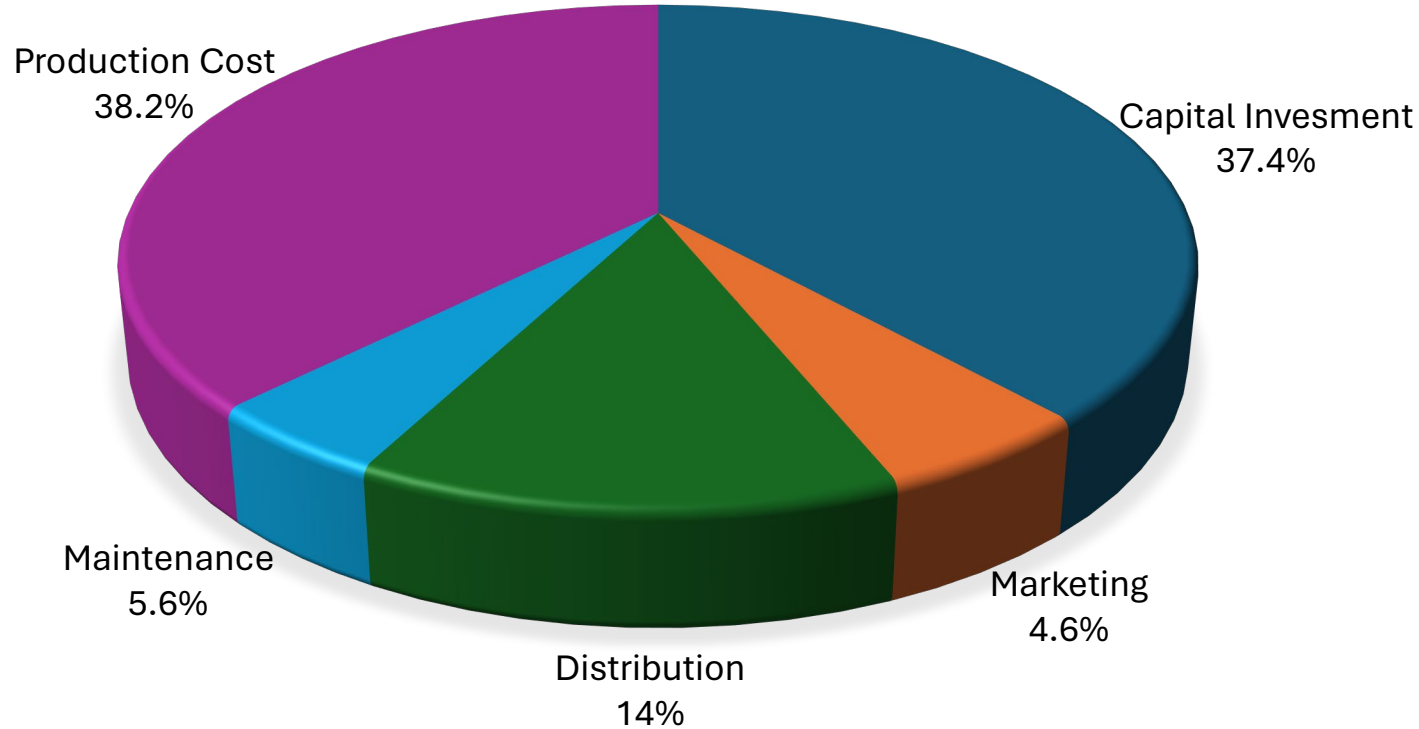
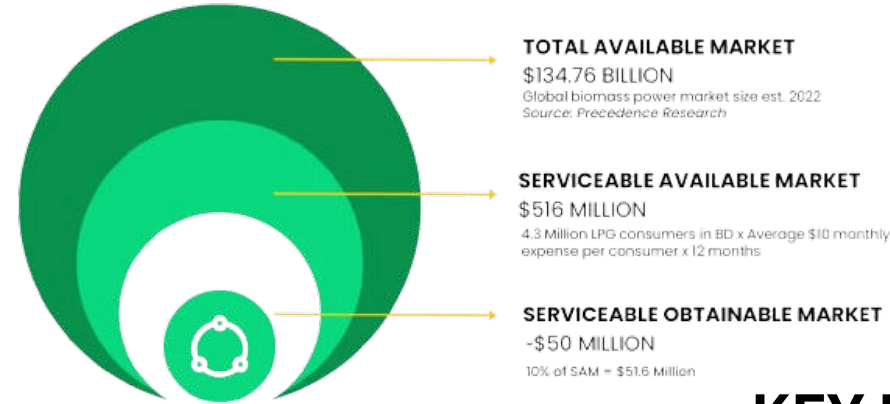
GOVERNMENT

Energy Import Dependence Reduced

Proof of Concept Renewable Sources

FINANCE

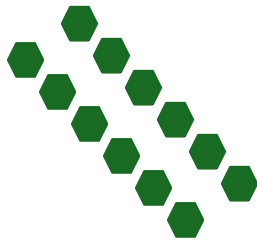
MARKET SIZE



KEY MILESTONES

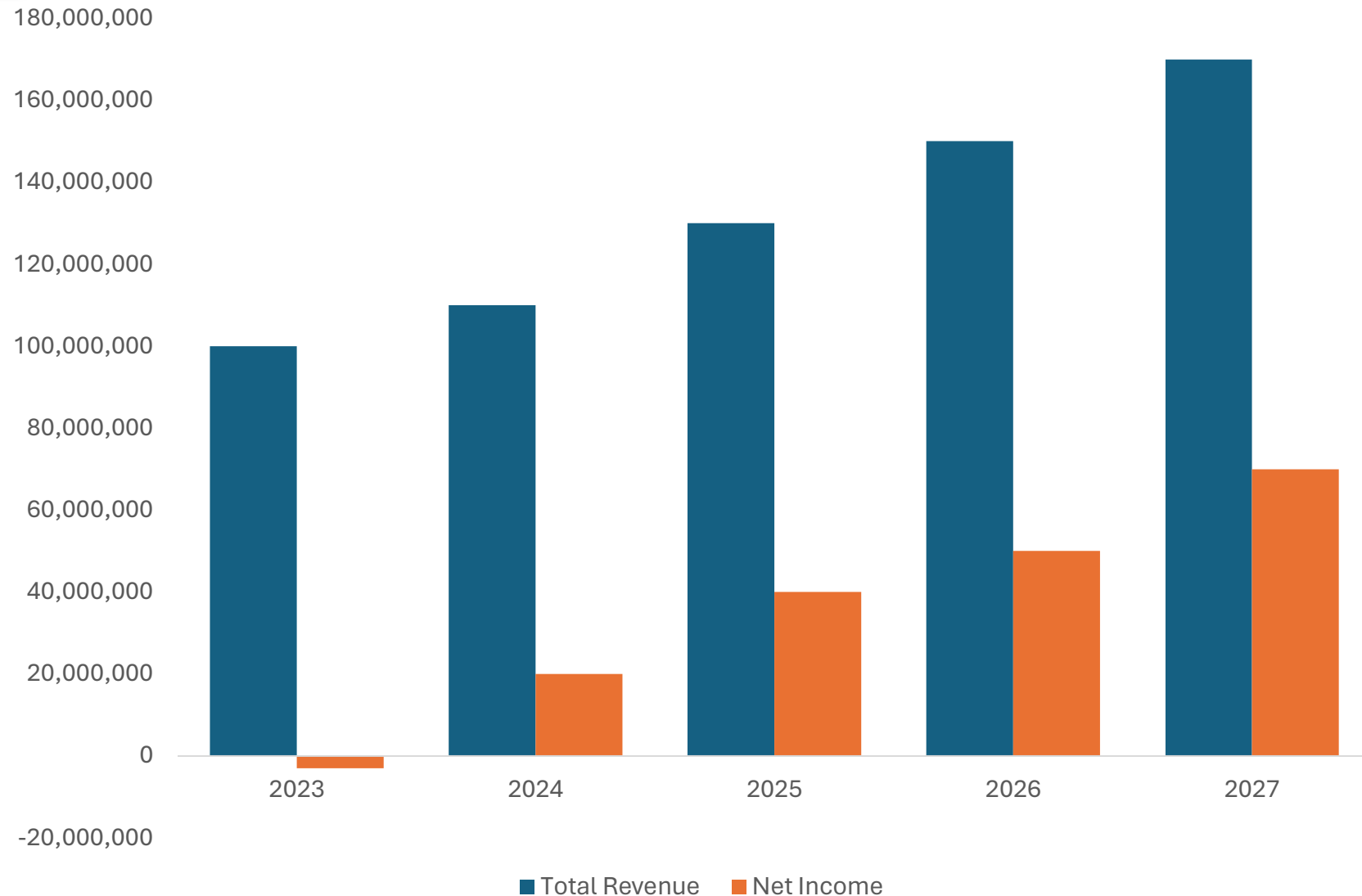
- CAPITAL INVESTMENT**
Facility 1 in year 1
Facility 2 in year 5
- DISTRIBUTION**
Expand outside Dhaka by year 3
- PRODUCTION**
Acquire 5% of the Domestic Cooking Gas Market
- MARKETING & SALES**
Sell 115k+ units (1.4M ltrs) by year 1

FINANCE

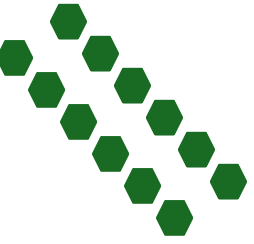


KEY INSIGHTS

- Payback Period**
3.96 years
- Assumed Growth**
Y1-3: 10% YoY
Y4-5: 15% YoY
- Price Increment**
Y3: Syngas (70->75)
Y4: Biochar (15->18)
- IRR 5Y**
12.38%



FEASIBILITY



BIO-GAS CYLINDERS

70 BDT/L at 56% Margin



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BDT 850

Small to Medium Households

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BDT 2,500

Medium to Large Households

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BDT 3,200

Restaurants/Hotels & Commercial Users

29% cheaper than LPG (1,176 BDT/ 12 LTR)



BIO-CHAR

By-Product of
Gasification Process

BDT 15/KG



Alternative revenue stream



Sold wholesale to farmers as
fertilizer



Create a 'green' process in
production



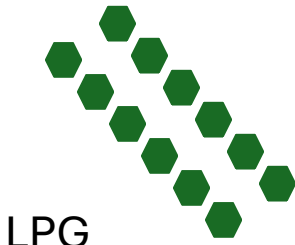
FEASIBILITY

END PRODUCT COMPARISON

To Alternative LPG Cylinder

Biogas is a more **Cost-effective renewable** for LPG gas for specific use-cases such as Cooking & powering Electricity Generators

	PRICE	LOW & STABLE	HIGH & VOLATILE
BIOGAS			
CARBON FOOTPRINT	LOW	LOW	
EFFICIENCY	HIGH	HIGH	
USAGE POTENTIAL	FOCUSED	VERSATILE	
RENEWABLE SOURE	YES	NO	



LPG



THANK YOU